Notes: Dittmar & Meisenzahl (RES, 2020)

Public Goods Institutions, Human Capital, and Growth: Evidence from German History

Contents

1 Big picture

- **Question.** What are the origins and consequences of the state as a provider of non-defense public goods?
- **Setting.** German cities during the Protestant Reformation. Some cities adopted church ordinances that transferred education and welfare provision to secular city governments.
- **Mechanism.** Public education created administrators who sustained the new public-goods equilibrium.
- **Main result.** Ordinance cities attracted and produced more upper-tail human capital and grew faster in the long run.
- **Identification.** The paper combines difference-in-differences with an IV based on early-1500s plague shocks.

2 Incremental contribution of the paper

- It identifies a non-military path to state capacity and public goods.
- It distinguishes formal public-goods institutions from Protestantism itself.
- It treats human capital as both an outcome of institutions and an input into state capacity.
- It constructs city-level upper-tail human-capital data from the Deutsche Biographie.
- It uses plague shocks as local political shocks interacting with the Reformation's global ideological shock.

3 Data and measurement

Church ordinances measure public-goods institutions. Upper-tail human capital is measured from the Deutsche Biographie. Population comes from Bairoch et al. Plague shocks come from historical plague chronologies. Publishing data come from the Universal Short Title Catalogue.

Detailed interpretation. The data link legal-institutional change, human-capital allocation, political competition, and long-run urban growth.

4 Models and empirical specifications

4.1 Difference-in-differences

$$Y_{it} = \alpha_i + \beta Post_t \times Law_i + \gamma_t + X_{it}'\delta + \epsilon_{it}.$$

Detailed interpretation. The coefficient on $Post \times Law$ measures whether adopting cities changed more after the Reformation than non-adopting cities.

4.2 Trend-shift model

$$Y_{it} = \alpha_i + \beta Post_t \times Law_i + \lambda PostTrend_t \times Law_i + \gamma_t + \epsilon_{it}.$$

Detailed interpretation. This separates immediate level effects from long-run trend changes.

4.3 Sectoral human-capital model

$$Y_{ist} = \alpha_i + \gamma_{st} + \beta_{s,t}(Law_i \times Period_t \times Sector_s) + controls + \epsilon_{ist}.$$

Detailed interpretation. The paper expects immediate effects in government, church, and education and later spillovers to business.

4.4 Cross-sectional long-run model

$$Y_i = c + \alpha Law_i + \gamma X_i + \epsilon_i.$$

Detailed interpretation. Outcomes include log population in 1800, upper-tail human capital in 1750–1799, and human-capital intensity.

4.5 Instrumental variables

$$Law_i = \pi Plagues_{i,1500-1522} + \Gamma X_i + u_i,$$
$$Y_i = \alpha \widehat{Law}_i + \Gamma X_i + \epsilon_i.$$

Detailed interpretation. Early-1500s plagues are used as local shocks to political equilibrium in the Reformation window.

5 Identification and empirical design

The difference-in-differences design compares adopting and non-adopting cities before and after the Reformation. Controls for Protestantism, region-year effects, principality-year effects, and pre-treatment outcomes distinguish public-goods institutions from Protestantism. The IV design uses plague shocks from 1500–1522 as political shocks that interacted with religious competition. Placebo windows, other institutional outcomes, and publishing responses support the exclusion restriction.

6 Section-by-section study guide

- **Introduction.** Explains the transition from minimal public goods to secular public-goods provision.
- **Historical background.** Describes church ordinances, schools, welfare, and political competition.
- **Data.** Introduces ordinances, Deutsche Biographie human capital, population, plagues, and publishing.
- **Human-capital results.** Shows migration and formation responses after adoption.
- **Long-run outcomes.** Shows population and human-capital growth by 1800.
- **IV analysis.** Uses plague shocks and provides exclusion-restriction tests.

7 Tables and figures

7.1 Figure 1: Jurisprudence and Construction + Figure 2: Religious Publishing

Read: Figure 1 shows jurisprudence publications and public-goods construction rising. Figure 2 shows religious publishing diverging in cities that later adopted ordinances.

Detailed interpretation. These figures motivate the mechanism: institutional change was preceded by legal knowledge, public-goods investment, and ideological competition.

Economic message. Legal and ideological activity supported the emergence of public-goods institutions.

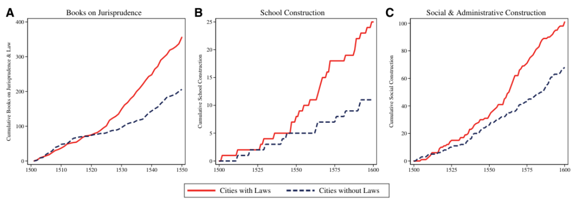

FIGURE 1
Jurisprudence and construction

(a) Figure 1: Jurisprudence and construction

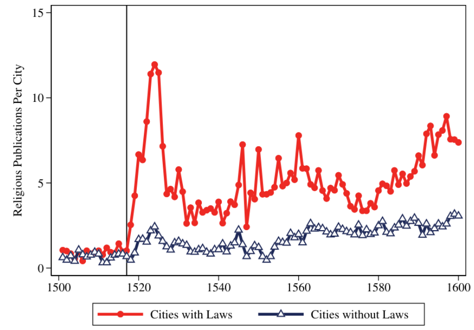

FIGURE 2
Religious publishing

(b) Figure 2: Religious publishing

7.2 Figure 3: Cities and Institutional Change + Figure 4: The Share of Cities Adopting Institutional Change

Read: Figure 3 maps ordinance adoption; Figure 4 shows adoption concentrated in the early 1500s.

Detailed interpretation. Adoption varied locally and was not simply universal Protestant diffusion.

Economic message. Institutional change was local and timed to the Reformation.

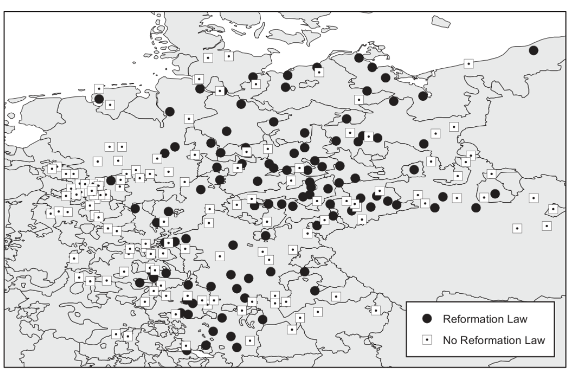

FIGURE 3
Cities and institutional change

(a) Figure 3: Cities and institutional change

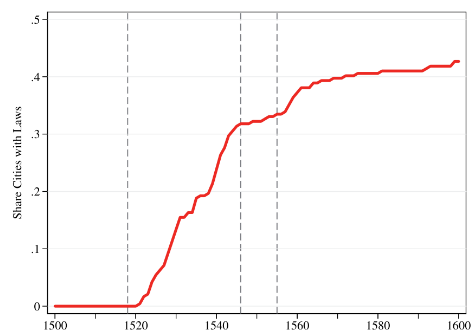

FIGURE 4
The share of cities adopting institutional change

(b) Figure 4: Share adopting

7.3 Table 1: Summary Statistics on Upper Tail Human Capital + Figure 5: The Migration of Upper Tail Human Capital

Read: After 1520, treated cities have much higher locally born, migrant, and total upper-tail human capital.

Detailed interpretation. This is descriptive evidence that ordinance cities became human-capital magnets and producers.

Economic message. Human capital is the mechanism linking institutions to growth.

TABLE 1
Summary statistics on upper tail human capital

	Cities with law			Cities without law			Difference
Upper tail human capital	N	Mean	Sd	N	Mean	Sd	HL statistic
Locally born pre-1520	103	1.02	3.37	136	0.20	0.64	0.00
Locally born post-1520	103	15.01	26.61	136	3.59	7.07	3.00***
Migrants pre-1520	103	0.41	0.86	136	0.13	0.67	0.00
Migrants post-1520	103	8.49	19.04	136	1.96	4.94	1.00
Total pre-1520	103	1.42	3.89	136	0.32	1.13	0.00
Total post-1520	103	23.50	44.89	136	5.54	11.55	5.00***

Notes: Upper tail human capital is measured by the number of people observed in the *Deutsche Biographie*. Locally born are people born in a given city i. Migrants to any given city i are individuals born in some other location j who died in city i. The last column presents the Hodges–Lehmann non-parametric statistic for the difference (median shift) between cities with laws and cities without laws. We use the Hodges-Lehmann statistic because we are examining non-negative distributions for which the standard deviation is larger than the mean and as a test statistic that is robust to outliers. Statistical significance at the 99%, 95%, and 90% levels denoted ***, **, and *, respectively.

(a) Table 1: Summary statistics

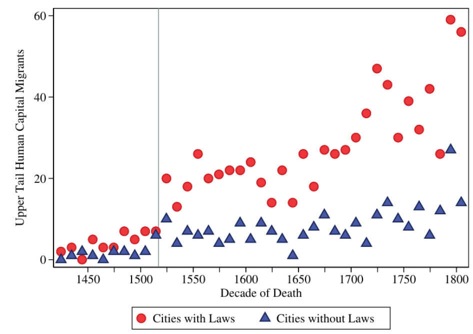

FIGURE 5
The migration of upper tail human capital

(b) Figure 5: Migration

7.4 Table 2: Public Goods Institutions and Upper Tail Human Capital

Read: Adoption raises migration and local formation of upper-tail human capital.

Detailed interpretation. The result holds with Protestantism controls and region/principality-year fixed effects.

Economic message. Public-goods institutions caused human-capital reallocation and production.

TABLE 2
Public goods institutions and upper tail human capital

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
Panel A: Log migration										
	Complete data: All cities 1420–1819				No religious heterogeneity	Complete data: All cities 1420–1819				No religious heterogeneity
Post × Law	0.22***	0.22***	0.26***	0.26**	0.30**	0.27***	0.33***	0.38***	0.35***	0.36***
	(0.06)	(0.06)	(0.07)	(0.11)	(0.15)	(0.07)	(0.07)	(0.08)	(0.10)	(0.13)
Post × Trend × Law						0.07**	0.09***	0.12***	0.11**	0.09
						(0.03)	(0.04)	(0.04)	(0.06)	(0.07)
Post × Protestant		−0.02	0.00	−0.05			−0.19***	−0.12	−0.13	
		(0.05)	(0.06)	(0.10)			(0.07)	(0.09)	(0.15)	
Post × Trend × Protestant							−0.08*	−0.08	−0.14	
							(0.05)	(0.06)	(0.11)	
Cities	239	239	239	234	167	239	239	239	234	167
Observations	1912	1912	1912	1872	1336	1912	1912	1912	1872	1336
R^2	0.59	0.59	0.66	0.72	0.73	0.65	0.65	0.71	0.76	0.80
Panel B: Log formation										
	Complete data: All cities 1420–1819				No religious heterogeneity	Complete data: All cities 1420–1819				No religious heterogeneity
Post × Law	0.33***	0.32***	0.37***	0.33***	0.33**	0.24***	0.26***	0.23***	0.24**	0.24*
	(0.06)	(0.07)	(0.07)	(0.11)	(0.15)	(0.06)	(0.07)	(0.08)	(0.10)	(0.12)
Post × Trend × Law						0.10***	0.11***	0.13***	0.08	0.07
						(0.04)	(0.04)	(0.05)	(0.06)	(0.06)
Post × Protestant		0.03	0.01	−0.05			−0.07	−0.12	0.03	
		(0.06)	(0.08)	(0.13)			(0.07)	(0.09)	(0.16)	
Post × Trend × Protestant							−0.03	0.07	0.06	
							(0.05)	(0.05)	(0.08)	
Cities	239	239	239	234	167	239	239	239	234	167
Observations	1912	1912	1912	1872	1336	1912	1912	1912	1872	1336
R^2	0.57	0.57	0.64	0.71	0.73	0.67	0.67	0.72	0.78	0.84
Controls										
City FE	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No
Trend	Yes	Yes	No	No	No	No	No	No	No	No
City-Specific Trend	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Region × Year FE	No	No	Yes	No	No	No	No	Yes	No	No
Principality × Year FE	No	No	No	Yes	Yes	No	No	No	Yes	Yes

Notes: This table presents regression estimates documenting the relationship between the adoption of public goods institutions and the migration and formation of upper tail human capital. Panel A studies the migration of upper tail human capital, measured by the log of the number of upper tail human capital migrants plus one. Columns 1–4 and 6–9 study the complete data 1420–1819. Columns 5 and 10 restrict analysis to religiously homogeneous principalities. Public goods institutions are measured by the adoption of a church ordinance (Law). The variable "Post × Law" is the interaction between indicators for the post-1520 period and for ever-treated status. "Post × Trend × Law" is the interaction between treated, post, and the linear time trend. "Protestant" is an indicator for Protestant cities from Cantoni (2012). Time periods are 50 year intervals: starting with 1420–1469 and ending with 1770–1819. Panel B studies the formation of upper tail human capital outcome in a city-period, measured by the log of the number of home-grown upper tail human capital individuals plus one. Region-year fixed effects interact region and year fixed effects, using 29 regions from Nüssli (2008). Principality-year fixed effects use 75 principalities (jurisdictions) recorded in the *Deutsches Städtebuch*. Standard errors in parentheses are clustered at the city level. Statistical significance at the 90%, 95%, and 99% confidence level denoted "*", "**", and "***".

Table 2: Public goods institutions and upper-tail human capital

7.5 Figure 6: Public Goods Institutions and Upper Tail Human Capital + Figure 7: Upper Tail Human Capital by Sector

Read: Figure 6 gives event-study evidence; Figure 7 decomposes migration and formation by sector.

Detailed interpretation. Effects first appear in government, church, and education, then spill over to business.

Economic message. The pathway is public-goods-sector human capital followed by private-sector spillovers.

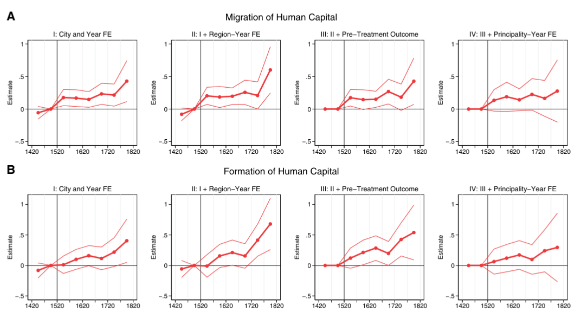

FIGURE 6

Public goods institutions and upper tail human capital

(a) Figure 6: Human capital

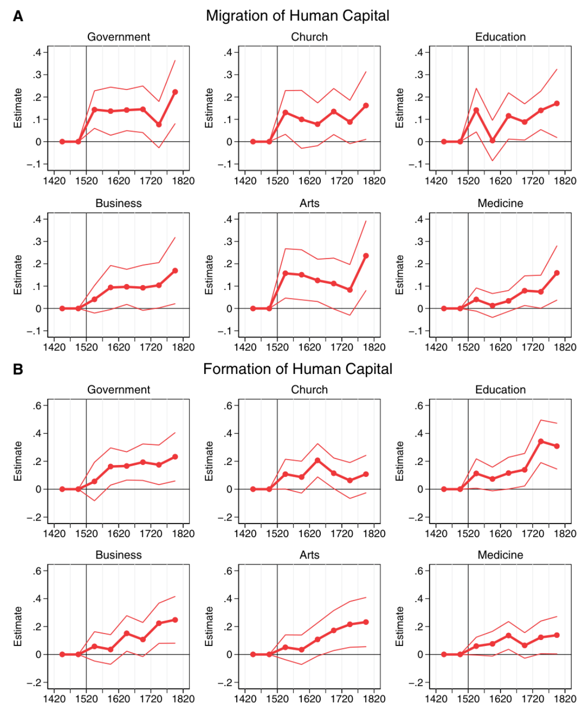

FIGURE 7

Upper tail human capital by sector. (A) Migration of human capital; (B) Formation of human capital

(b) Figure 7: Sectoral human capital

7.6 Tables 3–5: Population and Long-Run Outcomes

Read: Tables 3–5 show that ordinance cities became more likely to be observed as cities, had larger 1800 populations, and had more upper-tail human capital.

Detailed interpretation. The long-run effects are not driven by Protestantism alone.

Economic message. Public-goods institutions support long-run urban growth.

TABLE 3

Public goods institutions and city population in the panel

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
	Outcome: population observed Data on all towns				Outcome: population in top 100 Data on cities			
Post × Law	0.07***		0.07***	0.01	0.12**		0.13**	−0.10
	(0.02)		(0.02)	(0.03)	(0.05)		(0.05)	(0.09)
Post × Protestant		0.03**	0.02**	−0.02*		−0.01	−0.05	−0.06
		(0.01)	(0.01)	(0.01)		(0.06)	(0.06)	(0.12)
Post × Law × Trend				0.03**				0.12**
				(0.01)				(0.04)
Post × Prot × Trend				0.02**				0.00
				(0.01)				(0.06)
Observations	13,380	13,380	13,380	13,380	1,434	1,434	1,434	1,434
Town FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents regression estimates examining city population outcomes. In Columns 1–4, the outcome is a binary variable that takes the value of 1 if a town has population observed in Bairoch *et al.* (1988). Columns 1–4 examine the 2,230 towns recorded in the *Deutsches Städetebuch*. In columns 5–8 the outcome is a binary variable for cities with population in the top 100 and analysis is restricted to 239 cities observed in Bairoch *et al.* (1988). "Post × Law" interacts an indicator for institutional change and the post-1520 period. "Post × Protestant" interacts an indicator for cities that adopted Protestantism and the post-1520 period. Time trend interactions measure time in centuries from year 1500 (100 years is one time unit). We measure Protestantism using Cantoni (2012) for cities and the *Deutsches Städtebuch* for towns not in Cantoni (2012). Data on population are at 100-year intervals 1300 through 1800. Standard errors in parentheses are clustered at the town (city) level.

(a) Table 3

TABLE 4

Log population in 1800 by institutional change status and initial size

Population	[1] Cities with law	[2]	[3]	[4] Cities without law	[5]	[6]	[7] Difference	[8] Share	[9] Share
in 1500	N	Mean	Sd	N	Mean	Sd	in means	with law	of cities
Unobserved	35	1.81	0.43	94	1.59	0.50	0.22[0.01]	0.27	0.54
1–5 Thousand	32	2.01	0.61	30	1.69	0.57	0.32[0.04]	0.52	0.26
6–10 Thousand	20	2.37	0.85	8	2.50	0.59	−0.13[0.66]	0.71	0.12
11–20 Thousand	12	2.94	0.96	2	3.43	0.36	−0.49[0.26]	0.86	0.06
21+ Thousand	4	3.29	0.13	2	3.90	0.27	−0.61[0.16]	0.67	0.03
All cities	103	2.17	0.76	136	1.73	0.65	0.45[0.00]	0.43	1.00

Notes: This table presents the summary statistics for log city population in 1800 by institutional change status and initial pre-Reformation city size. Institutional change is measured by an indicator variable for whether a city had a church ordinance (Law) by 1600. Populations are measured in thousands: ln(*population*/1000). *P*-values for the statistical significance of differences in means in square brackets in column 7. Column 8 reports the share of cities with a church ordinance in each initial size category for population in 1500. Column 9 reports the share of total cities in each initial size category.

(b) Table 4

TABLE 5

OLS analysis of long-run outcomes

	[1]	[2]	[3]	[4]	[5]	[6]	[7]
Panel A: City population							
	Outcome: Ln Population in 1800						
Law	0.24*	0.26**	0.25**		0.28***	0.26***	0.37**
	(0.12)	(0.11)	(0.10)		(0.09)	(0.09)	(0.16)
Protestant				−0.08	−0.11	−0.12	
				(0.18)	(0.17)	(0.21)	
Observations	239	239	239	239	239	239	167
R^2	0.47	0.54	0.54	0.51	0.52	0.50	0.65
Panel B: Human capital							
	Outcome: Ln upper tail human capital 1750–1799						
Law	0.35***	0.41***	0.39***		0.30***	0.30***	0.39*
	(0.11)	(0.12)	(0.13)		(0.10)	(0.10)	(0.21)
Protestant				0.20	0.19	0.19	
				(0.20)	(0.22)	(0.22)	
Observations	239	239	239	239	239	239	167
R^2	0.29	0.40	0.40	0.34	0.35	0.35	0.60
Panel C: Human capital intensity							
	Outcome: Upper tail human capital per 1,000						
Law	0.09**	0.11***	0.11**		0.08*	0.08*	0.10
	(0.04)	(0.04)	(0.04)		(0.04)	(0.04)	(0.07)
Protestant				0.09	0.09	0.09	
				(0.08)	(0.08)	(0.09)	
Observations	239	239	239	239	239	239	167
R^2	0.25	0.42	0.42	0.30	0.31	0.30	0.59
Controls that vary across specifications							
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	No
Population FE	Yes	Yes	Yes	Yes	Yes	No	Yes
Main controls	No	Yes	Yes	No	No	No	No
Geo Controls	No	No	Yes	No	No	No	No
Cantoni Controls	No	No	No	Yes	Yes	Yes	Yes
Log Population 1500	No	No	No	No	No	Yes	No
Principality FE	No	No	No	No	No	No	Yes

Notes: This table presents the regression estimates of the relationship between the adoption of church ordinances (Law) and long-run outcomes. Outcomes are: in Panel A log population in 1800; in Panel B the log of upper tail human capital plus one 1750–99; and in Panel C upper tail human capital individuals 1750–99 per 1,000 population in 1800. Upper tail human capital is the sum of locally born individuals and migrants from the *Deutsche Biographie*. "Law" is an indicator. "Protestant" is an indicator for Protestant cities (Cantoni, 2012). All regressions control for Ln Upper Tail Human Capital in 1420–69 and 1470–1519. Columns 1–6 control for 29 region fixed effects using Nüssli (2008). Column 7 controls for 75 principality fixed effects using the *Deutsches Städtebuch*. "Main Controls" are as follows: Market rights by 1517; town incorporated by 1517; indicators for books printed pre-1517 (0, 1–100, 101–1000, 1001+), university by 1517; the number of students from each city receiving university degrees separately each 10-year period starting 1398 through 1508; and the average number of plagues from 1349 to 1499. "Geo Controls" are longitude, latitude, and their interaction. "Cantoni Controls" are year city founded and year turned Protestant, indicators for rivers, Hansa cities, Free-Imperial status, monasteries, university, printing, and distance to Wittenberg. Population fixed effects are indicators for 1500 population data: missing, 1,000–5,000, 6,000–10,000, 11,000–20,000, and 20,000+. Column 6 controls for log population in 1500, setting log population to 0 for cities with data unobserved, and an indicator for cities with data unobserved. Column 7 restricts analysis to religiously homogeneous principalities. Standard errors are clustered at the region (principality) level. Statistical significance at the 90%, 95%, and 99% confidence level denoted "*", "**", and "***".

(c) Table 5

7.7 Figure 8: City-Level Plague Outbreaks + Table 6: IV Analysis of Long-Run Outcomes

Read: Figure 8 shows plague timing; Table 6 shows a strong first stage and positive IV effects.

Detailed interpretation. The instrument uses local plague shocks in the Reformation window.

Economic message. Plagues mattered as political shocks, not merely as mortality shocks.

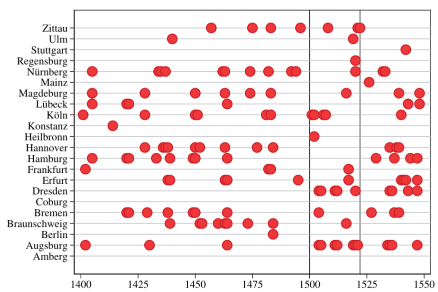

FIGURE 8

City-level plague outbreaks

(a) Figure 8

TABLE 6

IV analysis of long-run outcomes

	[1]	[2]	[3]	[4]	[5]	[6]
Panel A: first stage—public goods institutions						
	Outcome: adoption of law					
Plagues 1500–1522	0.14***	0.13***	0.13***	0.12***	0.12***	0.11***
	(0.02)	(0.02)	(0.03)	(0.02)	(0.04)	(0.03)
R^2	0.29	0.30	0.51	0.52	0.62	0.63
F Statistic on IV	34.02	32.69	18.92	31.42	8.70	12.02
Panel B: IV outcomes—population and human capital						
	Outcome: Ln Population in 1800					
Law	1.61**	1.76**	1.94**	2.17**	2.13*	2.30**
	(0.82)	(0.83)	(0.98)	(0.93)	(1.12)	(1.01)
	Outcome: Ln upper tail human capital 1750–1799					
Law	2.74**	2.96**	3.10**	3.34**	4.23**	4.54***
	(1.24)	(1.29)	(1.38)	(1.40)	(1.81)	(1.28)
	Outcome: upper tail human capital per 1,000					
Law	0.56**	0.63**	0.59*	0.66**	0.80**	0.92***
	(0.27)	(0.25)	(0.31)	(0.29)	(0.40)	(0.29)
Controls						
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Plagues pre-1500: level	Yes	Yes	Yes	Yes	Yes	Yes
Plagues pre-1500: non-linear	No	Yes	No	Yes	No	Yes
Region fixed effect (n=29)	No	No	Yes	Yes	No	No
Principality fixed effect (n=75)	No	No	No	No	Yes	Yes
Observations	239	239	239	239	234	234

Notes: The first-stage outcome variable in Panel A is an indicator for the adoption of a church ordinance (Law). "Plagues 1500-1522" is the number of plagues 1500–1522. The outcome variables in Panel B are: log population in 1800; log of the number of upper tail human capital individuals observed between 1750 to 1799 plus one; and the number of upper tail human capital individuals per thousand population. In first-stage regressions, the dependent variable is an indicator for the passage of a Reformation ordinance by 1600. All regressions control separately for the log of upper tail human capital observed 1420–1469 and in 1470–1519, and include the complete set of controls from Table 5. Upper tail human capital is measured by the sum of the number of migrants dying in a city-period and the number of people locally born people reaching age forty in a city-period. Region fixed effects control use territories recorded by Nüssli (2008). Principality fixed effects use principalities recorded in the *Deutsches Städtebuch*. "Plagues Pre-1500: Level" is the average number of plagues from 1350 to 1499. "Plagues Pre-1500: Non-Linear" indicates independent controls for the number of years with plague outbreaks in each of the twenty-five year periods: 1350–74 through 1475–99. Standard errors are clustered at the 1500 Region level. Statistical significance at the 90%, 95%, and 99% confidence level denoted "*", "**", and "***".

(b) Table 6

7.8 Tables 7–10: Exclusion-Restriction and Mechanism Tests

Read: Table 7 tests plague windows; Table 8 tests other institutions; Table 9 tests publishing; Table 10 tests alternative plague timing.

Detailed interpretation. These tests support the claim that early-1500s plagues shifted political competition and adoption rather than directly causing growth.

Economic message. The IV mechanism is Reformation-era political competition.

TABLE 7
Historic plague outbreaks and city population in 1800

	[1]	[2]	[3]	[4]
	Outcome: Ln population in 1800			
Plagues 1350–1374	0.42***	0.29***	0.29***	0.27***
	(0.05)	(0.08)	(0.08)	(0.09)
Plagues 1375–1399	0.07	0.12	0.11	0.12
	(0.10)	(0.12)	(0.12)	(0.15)
Plagues 1400–1324	−0.07	−0.08	−0.08	−0.06
	(0.19)	(0.22)	(0.22)	(0.26)
Plagues 1425–1449	0.11	0.13	0.16*	0.13
	(0.09)	(0.08)	(0.08)	(0.09)
Plagues 1450–1474	−0.00	0.01	0.01	0.02
	(0.11)	(0.09)	(0.09)	(0.09)
Plagues 1475–1499	0.20	0.13	0.13	0.10
	(0.23)	(0.20)	(0.20)	(0.19)
Plagues 1500–1524	0.19**	0.20**	0.20**	0.22**
	(0.07)	(0.07)	(0.08)	(0.09)
Plagues 1525–1549	0.03	0.03	0.02	0.02
	(0.07)	(0.06)	(0.07)	(0.08)
Plagues 1550–1574	0.10	0.09	0.10	0.08
	(0.07)	(0.05)	(0.06)	(0.08)
Plagues 1575–1599	0.00	−0.06	−0.06	−0.04
	(0.05)	(0.06)	(0.06)	(0.07)
Observations	239	239	239	239
Population in 1300	No	Yes	Yes	Yes
Controls	No	No	Yes	Yes
Region fixed effects	No	No	No	Yes

Notes: This table presents results from regressions estimating the relationship between city population in 1800 and historic plague exposure between 1350 and 1599. "Plagues 1350-1374" is the count of plague outbreaks in that period. Other plague variables are similarly defined. Controls include indicators for city incorporation and for city market rights granted by 1300. Population fixed effects are for categorical variables: population in 1300 data missing; 1,000–5,000; 6,000–10,000; 11,000–20,000; and more than 20,000. Region fixed effects control for regions from the *Deutsches Städtebuch*. Standard errors are clustered at the region level. Statistical significance at the 90%, 95%, and 99% confidence level denoted "*", "**", and "***".

(a) Table 7

TABLE 8
Plagues and other potential institutional changes in the 1500s

	[1]	[2]	[3]	[4]	[5]	[6]
	Outcome: change in institutions in the 1500s					
	Participatory elections		Public representation		Guild constitution	
Plagues 1500–1522	−0.01	−0.01	0.03	0.02	−0.07	−0.08
	(0.02)	(0.02)	(0.03)	(0.03)	(0.05)	(0.06)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	No	Yes	No	Yes	No	Yes
Observations	239	239	239	239	239	239

Notes: This table presents results from regressions estimating the relationship between plague shocks and institutional change in the 1500s. We examine changes in formal institutions establishing or eliminating: (1) participatory elections; (2) public representation, specifically formal representative bodies for citizens (*Burghers*); and (3) constitutions securing guild representation on city councils. The outcome measures of institutional change are constructed from data in Wahl (2016, 2017) and the *Deutsches Städtebuch*. Institutional change outcomes take values of: 1 when institutional indices go from 0 in 1500 to 1 in 1600; 0 when institutional indices do not change between 1500 and 1600; and −1 when institutional indices go from 1 in 1500 to 0 in 1600. Specifications and the complete set of controls match the first-stage regression estimates reported in Table 6. Standard errors are clustered by region.

(b) Table 8

TABLE 9
Plagues and religious and political publications

	[1]	[2]	[3]	[4]
	Religious publications		Political publications	
Post × Plagues 1500–1522	2.03***	2.04***	0.09***	0.09***
	(0.45)	(0.50)	(0.02)	(0.02)
City fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	No	Yes	No
Region × year fixed effects	No	Yes	No	Yes
Observations	24,139	24,139	24,139	24,139
Mean outcome	1.46	1.46	0.04	0.04

Notes: This table presents regression estimates examining the relationship between plagues and religious and political publishing outcomes. The outcome is the number of religious or political publications in a given city-year. Data on religious and political publications are from the *Universal Short Title Catalogue*. The analysis examines printing across cities from 1500 through 1600. "Post × Plagues 1500-1522" is an interaction between an indicator for the post-1517 period (Post) and the number of plagues in the early 1500s (Plagues 1500–1522). Standard errors are clustered by city. Statistical significance at the 90%, 95%, and 99% confidence level denoted "*", "**", and "***".

(a) Table 9

TABLE 10
IV Estimates using plague shocks in different time periods

	[1]	[2]	[3]	[4]
	IV outcome: Ln population in 1800			
Law	1.76**	2.17**	1.33*	1.96**
	(0.83)	(0.93)	(0.77)	(0.81)
F Statistic on IV	32.69	31.42	31.94	28.10
	First-stage outcome: law			
Plagues 1500–1522	0.13***	0.12***		
	(0.02)	(0.02)		
Plagues 1500–1550			0.19***	0.15***
			(0.03)	(0.03)
Observations	239	239	239	239
Region fixed effects	No	Yes	No	Yes

This table presents IV regression estimates. Columns 1 and 2 present our baseline estimates using plague 1500–1522 as the IV. Columns 3 and 4 present corresponding estimates using plague 1500–1550 as the IV. For the period 1500–1550, we normalize the count of plagues to a 23 year basis to ensure comparability of the first-stage estimates (*i.e.* for Plagues 1500–1550 we divide the observed number of plagues by 51 and multiply by 23). Regression specifications and controls correspond to Table 6 column 2 (without region fixed effects) and column 4 (with region fixed effects). Standard errors are clustered by region. Statistical significance at the 90%, 95%, and 99% confidence level denoted "*", "**", and "***".

(b) Table 10

8 Short summary

- The paper studies how German cities shifted to public-goods institutions during the Reformation.
- Church ordinances created public education, welfare, and administration.
- Adopting cities attracted and produced more upper-tail human capital.
- The first human-capital effects appear in government, church, and education, with later business spillovers.
- Long-run population and human-capital outcomes are higher in adopting cities.
- Early-1500s plague shocks predict adoption and support a causal interpretation.
- Placebo windows, other-institution tests, and publishing evidence support the political-competition mechanism.

9 Conclusion

9.1 Core conclusion

Public-goods institutions created during the Reformation increased human capital and long-run urban growth.

9.2 Incremental contribution

The paper identifies a non-military path to state capacity and shows that human capital is both an input into and outcome of public-goods institutions.

9.3 Economic intuition

Public education produced administrators who sustained the new equilibrium, while better administration and schooling later spilled over into business and city growth.

9.4 Limitations

The evidence is historical, the human-capital measure focuses on elites, and the IV identifies cities whose adoption was shifted by plague shocks.

9.5 Takeaway

Public-goods institutions and human capital can become mutually self-reinforcing, generating persistent long-run development effects.